

Pavan Muguda Sanjeevamurthy

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SUMMARY

Remote Sensing Data Scientist and PhD researcher with 3+ years of experience in developing production-grade Earth Observation data processing pipelines. Expertise in SAR/InSAR processing, soil moisture retrieval algorithms, and machine learning for geospatial applications. Proven track record of delivering operational products within ESA Copernicus and EUMETSAT H SAF programmes.

EDUCATION

PhD in Remote Sensing (ongoing)

TU Wien, Department of Geodesy and Geoinformation

Oct 2022 – Present

Vienna, Austria

M.Sc. in Earth Oriented Space Science and Technology (ESPACE)

Technical University of Munich, Aerospace Geodesy
Specialisation: Remote Sensing

Oct 2019 – May 2022

Munich, Germany

B.E. in Mechanical Engineering

Nitte Meenakshi Institute of Technology

Jun 2013 – Apr 2017

Bengaluru, India

TECHNICAL SKILLS

Languages	Python, MATLAB, IDL, C, Julia
ML / DL	PyTorch, segmentation_models_pytorch, ClearML, scikit-learn
Remote Sensing	ESA SNAP, Google Earth Engine, ISCE, PyGMTSAR, MintPy
Geospatial Stack	xarray, rioxarray, GeoPandas, Dask, pyproj, GDAL
DevOps / Infra	Linux, Docker, Git, GitLab CI/CD, HPC (SLURM)
GIS	ArcGIS, QGIS
Spoken Languages	English, Kannada, Hindi, Telugu

PROFESSIONAL EXPERIENCE

Data Scientist – Remote Sensing

Space42

Oct 2025 – Present

Abu Dhabi, UAE

- Developing SAR and InSAR data processing pipelines for operational geospatial intelligence products.
- Building deep-learning models for building-height estimation from very-high-resolution SAR imagery.
- Designing and maintaining SBAS InSAR pipelines for surface deformation monitoring.

Project Assistant (Research)

TU Wien, Department of Geodesy and Geoinformation

Oct 2022 – Sep 2025

Vienna, Austria

CGLOPS – Copernicus Global Land Operations Service (Soil Moisture)

- Built a near-real-time pipeline fusing Sentinel-1 SAR and ASCAT surface soil moisture to produce SCATSAR Soil Water Index at 0.5 km resolution for Europe.
- Developed core libraries implementing recursive and weighted-recursive exponential filter formulations for soil water index generation.
- Created a global NRT pipeline generating Soil Water Index at 12.5 km resolution from ASCAT SSM data.
- Conducted quality assurance of SCATSAR and SCAT SWI products using in-situ and modelled root-zone soil moisture.
- Contributed to ATBD, Product Validation Report, and Product User Manual documentation.

ROSSIHNI – Remote Sensing and Social Interest for Humanitarian Insights

- Developed a drought indicator workflow for MetOp ASCAT SSM at 6.25 km for climate data records and near-real-time applications over East Africa.
- Performed quality assurance and assessed soil moisture, vegetation, and drought seasonal patterns using Sentinel-1 and ASCAT products.

Clim4Cast

- Refactored pipelines for generating Sentinel-1-based downscaling parameters.
- Produced and validated MetOp ASCAT disaggregated surface and root-zone soil moisture at 0.5 km resolution for Europe.

Additional: Conducted a climate use case investigating monsoon intensification in India.

Research Assistant

Universität der Bundeswehr München

Aug 2022 – Nov 2022

Munich, Germany

- Developed a self-supervised learning framework for InSAR phase and coherence estimation.

- Built a pre-processing pipeline for Persistent Scatterer Interferometry from Sentinel-1 data.

Student Research Assistant

Technical University of Munich

Feb 2022 – Jun 2022

Munich, Germany

- Developed algorithms for filtering buildings from 3D city models and computing shadow volumes and occlusions.

Master Thesis: InSAR Denoising using Self-Supervised Learning

Data Science in Earth Observation, TUM

Nov 2021 – May 2022

Munich, Germany

- Designed a complex-valued U-Net with non-local attention blocks for denoising InSAR phase data.
- Built simulation pipelines for generating SAR interferograms and training datasets.
- Benchmarked proposed model against state-of-the-art InSAR filters on simulated and real data.

Student Assistant – SAR Signal Processing

German Aerospace Center (DLR)

Apr 2020 – Sep 2021

Oberpfaffenhofen, Germany

- Developed a standalone Python GUI for extracting and visualising SAR image chips from Sentinel-1 SLC data.
- Ported the ESD co-registration algorithm from IDL to Python.
- Built SNAP plugins and Python visualisation libraries for multitemporal SAR/InSAR analysis.

SELECTED PUBLICATIONS

- S. Massart, M. Vreugdenhil, S. Hahn, **P. M. Sanjeevamurthy**, C. Villegas-Lituma, W. Wagner. “High-resolution drought monitoring with Sentinel-1: A case-study over Mozambique.” *Agricultural Water Management*, 318(109638), 2025.
- F. Sica, **P. M. Sanjeevamurthy**, M. Schmitt. “Self-Supervised Learning for InSAR Phase and Coherence Estimation.” *IGARSS 2023*, IEEE, 2023.
- **P. M. Sanjeevamurthy**, M. Vreugdenhil, S. Hahn, S. Massart, C. Villegas-Lituma, R. Lindorfer, W. Wagner. “A Case Study on Agricultural Drought Monitoring using ASCAT SSM at 6.25 km over Eastern Africa.” *EGU General Assembly*, EGU24-6782, 2024.
- **P. M. Sanjeevamurthy**, S. Hahn, M. Vreugdenhil, B. Raml, W. Wagner. “H SAF ASCAT Disaggregated SSM at 0.5 km: First Validation Results with ISMN Data Over Europe.” *Living Planet Symposium*, 2025.

Additional conference contributions at EGU 2023 as co-author.

HONOURS

Deutschlandstipendium Scholarship

Federal Government of Germany merit scholarship for academic excellence.

Oct 2019 – Sep 2020